

CPS 150 Review Session #4

Conditional Logic & Loops

Review Activity

What would this code snippet output?

```
int x = 10;

if (x > 5) {
    if (x < 15) {
        System.out.println("A");
    } else {
        System.out.println("B");
    }
} else {
    System.out.println("C");
}
```

Character & String Methods

What does this snippet output?

```
String myString = "Hello World";  
char myChar1 = myString.charAt(5);  
char myChar2 = myString.charAt(7);  
  
if (!Character.isWhitespace(myChar1)) {  
    System.out.println("Condition 1 is true");  
}  
  
if (!Character.isWhitespace(myChar2)) {  
    System.out.println("Condition 2 is true");  
}
```

Conditional Expressions (Ternary Operator)

Allows us to quickly do an if-else check within a singular line. Good for conditional variable assignment.

```
String numType = (num % 2 == 0) ? "Even" : "Odd";
```

Comparing Floating Point Numbers

Floating point numbers (`double` , `float`) are not perfectly precise, making them hard to compare. The common work is to take a difference and compare it to a negligibly small number.

```
double num1 = 30.3;
double num2 = 10.1 + 20.2; // 30.299999999999997
double epsilon = 1e-8;

if (num1 == num2)
    // will not execute because 30.3 != 30.299999999999997
if (Math.abs(num1 - num2) < epsilon)
    // will execute
```

Looping

What would each variables equal to stop the condition?

```
while (x < 20)
```

```
while (y >= 1)
```

```
while (!input.equals('quit'))
```

```
while (input2 != -1)
```

```
String myString = "Hello World";
```

```
while (z < myString.length())
```

Looping

What will the output of this snippet be?

```
int x = 1;
int y = 7;

while (x < y) {
    System.out.println(x + " " + y);

    x += 2;
    y--;
}
```

Practice Activity

Write a program that finds the average of all the positive numbers entered by the user. The user input will stop on entering a negative number.

```
Enter a list numbers to be averaged. A negative number will stop input.  
4  
6  
3  
2  
8  
7  
-1  
The average is 5.0
```